Evaluation of controls on silicate weathering in tropical mountainous rivers: Insights from the Isthmus of Panama

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ABSTRACT

The Isthmus of Panama comprises a lithologically diverse andesitic oceanic arc of Late Cretaceous to Holocene age; it has large spatial variation in rainfall, displays a large range of physical erosion rates, and, therefore, is an ideal location to examine silicate weathering in the tropics. We use a multiyear data set of river chemistry for a 450 km transect across the Cordillera Central of west-central Panama to investigate controls on chemical weathering in tropical small mountainous rivers. Sea-salt corrected cation weathering yields (Ca_{sil} + Mg_{sil} + Na + K) range over more than an order in magnitude from 3.1 to 31.7 t/km^2/yr, while silicate weathering yields (Ca_{sil} + Mg_{sil} + Na + K + Si) range from 6.9 to 69.5 t/km^2/yr. Watershed lithology is the primary control on riverine chemistry, but landscape topographic character and land cover and/or land use also influence solute delivery potential. Strong statistical links of small mountainous river chemical weathering fluxes with rainfall and physical weathering rates attest to the importance of runoff and erosion in maintaining elevated bedrock weathering rates. CO_2 consumption ranges from 155 × 10^3 mol/km^2/yr to 1566 × 10^3 mol/km^2/yr, in the upper range of global rates, leading us to suggest that andesite terrains should be considered separately when calculating removal of CO_2 from the atmosphere via silicate weathering.

INTRODUCTION

Atmospheric CO_2 consumption via silicate weathering is an important control on Earth's climate over million-year time scales (Berner et al., 1983). From the pioneering work of Garrels and Mackenzie (1971) and Berner et al. (1983) onward, many attempts have been made to identify potential controls of silicate weathering rates and associated CO_2 drawdown at local, regional, and global scales. A study to quantify CO_2 drawdown via silicate weathering from the world's 60 largest rivers by Gaillardet et al. (1999) demonstrated that a disproportionate quantity of silicate weathering occurs in the topics, largely as a function of elevated runoff and temperature; it was noted that these high chemical denudation rates are often maintained by correspondingly high physical erosion rates.

This linkage of physical and chemical weathering rates was reinforced by subsequent studies at active plate margins (Jacobson and Blum, 2003; Lyons et al., 2005; Gaillardet et al., 2011), where the majority of the physical erosion on Earth's surface currently occurs. It has been hypothesized that high physical erosion rates continuously provide abundant mineral surfaces and/or maintain an ideal soil thickness for uninterrupted chemical weathering (West, 2012). The role of uplift and erosion in maintaining high chemical weathering rates has been validated through numerical modeling (Hilley and Porder, 2008; Hartmann et al., 2009). Therefore, the humid tropics should provide an ideal location to investigate the physical controls and role of land cover and landscape alteration practices, such as deforestation and agriculture, on silicate weathering rates. While previous studies have linked land-clearing practices with increased discharge and sediment yields in the tropics (Syvitski and Milliman, 2007), the link to silicate weathering rates remains unexplored.

Much attention has been given to the weathering of mafic silicates and their potential for CO_2 drawdown, e.g., for the Deccan Traps of India at the regional scale (Dessert et al., 2001), or at a small spatial scale on volcanic islands such as Réunion (Louvat and Allègre, 1997). These efforts led to compilation studies for basalt terrains that identified CO_2 drawdown values rates as high as 30%–35% of that for granites, despite their limited aerial extent (Gaillardet et al., 1999; Dessert et al., 2003). Subsequent investigations of andesitic-dacitic lithologies confirmed their enhanced weathering potential (Rad et al., 2006; Goldsmith et al., 2010) relative to granitic terrains, albeit to a lesser extent (Gurumurthy et al., 2012).

The Isthmus of Panama is situated between 7°N and 10°N and 77°W and 83°W, (Fig. 1), and comprises a variety of igneous lithologies ranging from mafic to intermediate and evolved compositions; it receives a large north-south spatial variation in rainfall of 1300 to >4000 mm/yr and has a wide range of physical erosion rates of 26–597 m/m.y. (Sosa-Gonzales, 2011). Thus, Panama is an exemplary location to investigate controls on silicate weathering in the humid tropics. Harmon et al. (2009), in a study within the Rio Chagres watershed in the Panama Canal region, observed a chemical weathering rate of 10^8 t/km^2/yr, which is well above the global average and appears to be sustained by correspondingly high rates of physical erosion (275–289 t/km^2/yr; Nichols et al., 2005). However, it is unclear whether these chemical weathering rates are the result of idealized local conditions or indicative for the small mountainous rivers of Panama. Here we evaluate a multiyear data set of large spatial extent to evaluate controls on chemical weathering across west-central Panama.

STUDY AREA

The landscape that today composes the Isthmus of Panama (Fig. 1) formed over the past 100 m.y. as a consequence of Caribbean, South American, Cocos, and Nazca plate interactions that generated an island arc archipelago on Mesozoic oceanic crust of the Caribbean plate (Molnar and Sykes, 1969). Geologically, Panama is a mosaic of bedrock lithologies including hydrothermally altered marine oceanic crust, intrusive igneous rocks of mafic to felsic composition, and an array of submarine and subaerial volcanics of tholeiitic, low-K calc-alkaline, and adaktic affinity. Loosely consolidated volcaniclastics and shallow-marine sediments were deposited unconformably upon the igneous rocks of the Tertiary volcanic arcs in the Bocas del Toro region, the Burica and Azuero Peninsulas, and in portions of the Colon, Trans-Isthmus, and east Panama regions (Wegner et al., 2011).

METHODOLOGY

This work is part of a larger geochemical study of streams and rivers in 78 watersheds across west-central Panama undertaken from 2005 to 2012. Cation and silicate fluxes and CO_2 consumption values were calculated using a subset consisting of 62 samples from two regions: 35 grab samples from 5 rivers of the greater Panama Canal watershed collected during 2005–2009, and 27 samples from 14 rivers across west-central Panama collected during 2006–2007. The data set was corrected for precipitation (after Murphy and Stallard, 2012) and for the nonsilicate contribution of Ca and Mg

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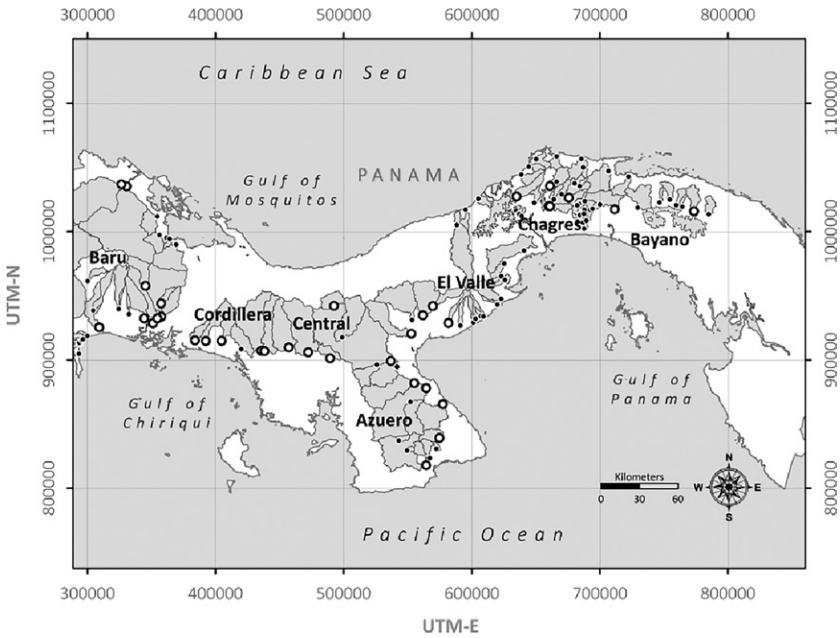

Figure 1. Outline map of Panama showing the 78 watersheds sampled. UTM coordinates. See text for discussion. For a list of rivers and their major tributaries sampled in this study, see Table DR1 (see footnote 1). Black circles denote locations where fluxes were calculated as part of this study. Small black dots denote locations where water samples were collected as part of a larger study.

using the andesite end-member ratios of 0.5 for Ca/Na and Mg/Na of Gaillardet et al. (1999) prior to its use in chemical denudation and CO_2 consumption calculations. A multistep process was employed whereby individual cation compositions were first multiplied by the average daily discharge value from gauging stations of either the Empresa de Transmisión Eléctrica or Autoridad del Canal de Panama to produce an instantaneous denudation flux. Plots comparing the instantaneous denudation fluxes with respective discharge values were then compiled to produce specific elemental flux determination equations. High r^2 values support this approach (Ca = 0.88; Mg = 0.92; Na = 0.92; K = 0.80; Si = 0.96). Long-term monthly discharge values were substituted into the equations and calculated denudation values divided by watershed area to produce cation and silicate weathering yields for 36 rivers. CO_2 consumption was calculated by dividing the precipitation and silicate corrected Na + K + Mg(2x) + Ca(2x) annual flux values by watershed area.

The influences of geomorphology, land use, land cover, and climate on weathering fluxes were analyzed using three widely available geospatial data sets. The National Aeronautics and Space Administration Jet Propulsion Laboratory's Shuttle Radar Topography Mission (SRTM) 90 m resolution digital elevation model (www2.jpl.nasa.gov/srtm/) was used to estimate watershed areas and stream gradients. Average annual low and high temperatures and annual minimum and maximum precipitation values were determined for each watershed using data obtained from the ESRI ArcAtlas (gcmd.nasa.gov/records/GCMD_ESRI_Arc_Atlas.html). Land use and land cover fractions for each watershed were estimated using data provided by the European Space Agency 2009 Global Land Cover Map (due.esrin.esa.int/page_globcover.php).

RESULTS

Ranges of in situ measurements of water temperature (T), electrical conductivity (specific conductance, SPC), and pH for more than 858 samples from 78 rivers and streams across west-central Panama are T = 12.7–34.6 °C, SPC = 145–2924 mS/cm, and pH = 4.9–8.7. In general, Panamanian waters are dilute, with an average concentration of total dissolved solids of 136 ± 139 mg/L and total cation content of T_{z+} = 1.35 ± 1.189 meq/L. Overall, the weathering of igneous rock, dominated by plagioclase dissolution, produces stream and rivers waters characterized by Ca > Na > Mg > K and HCO_3 > Cl > SO_4 > NO_3. Cation weathering fluxes (Ca_{sil} + Mg_{sil} + Na + K) range over more than an order in magnitude from 3.1 to 31.7 t/km^2/yr, with the highest regional average values identified for the Chiriquí and Bocas del Toro regions, 22.5 t/km^2/yr and 21.4 t/km^2/yr, respectively (Table 1; Table DR3 in the GSA Data Repository[1]). The range of variation for silicate weathering yields was slightly larger, from 6.9 to 69.5 t/km^2/yr, and exhibited similar regional trends (Chiriquí region, 49.3 t/km^2/yr; Bocas del Toro region, 46.8 t/km^2/yr). Associated CO_2 consumption rates are between 155 × 10^3 mol/km^2/yr and 1566 × 10^3 mol/km^2/yr. As noted in Table 2, a Pearson statistical analysis between our calculated cation and silicate weathering fluxes and climatic, land use, land cover, and geomorphological parameters revealed significant correlations that vary in strength and direction.

DISCUSSION

Although Panama silicate weathering yields (Table 1) are well below those determined for watersheds draining ultramafic-dacitic arc rocks on the island of Luzon in the Philippines (Schopka et al., 2011), they are similar to those observed for the Lesser Antilles (Rad et al., 2006; Goldsmith et al., 2008, 2010), and thus confirm the weathering potential of andesitic terrains. CO_2 consumption rates for Panama approach the higher values observed in other actively eroding tectonically convergent regions such as the Himalayas (Galy and France-Lanord, 1999) and are in the low to middle range determined for the andesitic volcanic settings of Martinique and Guadalupe (Rad et al., 2006) and Dominica (Goldsmith et al., 2010). That these CO_2 consumption rates are at the lower end of those determined for the basaltic terrains of Réunion (Louvat and Allègre, 1997) and the Deccan Traps (Dessert et al., 2001) may be a reflection of a comparatively lower runoff in Panama coupled with lithology-limited reaction kinetics.

Regionally, the highest weathering fluxes in Panama occur in the high mountains of the Cordillera Central, the Chagres uplands, and the Bayano region (Fig. 1). This finding supports the role of orographic-driven precipitation on elevated weathering rates observed in Guadeloupe (Gaillardet et al., 2011). The importance of runoff on weathering rates has been well documented through worldwide river compilation studies and modeling studies (Meybeck, 1987; Gaillardet et al., 1999). The strong significant correlation between weathering fluxes and runoff in Panama is not surprising given our flux calculation methodology. However, this finding was independently confirmed by the positive significant correlations between weathering fluxes and minimum and maximum precipitation. Of the top 10 silicate weathering yields, seven were observed in the Chiriquí region; this is probably a reflection of the high inputs of Si from the more readily erodible volcaniclastics of the Quaternary age Volcan Baru.

The correlation of weathering rates with long-term physical erosion rates is observed despite differences in time scales for the respective calculation techniques. This suggests that Panama's geologically recent uplift history has had a continuing impact on both localized topography and associated rainfall and, by extension, overall

[1]GSA Data Repository item 2015195, Tables DR1–DR4, is available online at www.geosociety.org/pubs/ft2015.htm, or on request from editing@geosociety.org or Documents Secretary, GSA, P.O. Box 9140, Boulder, CO 80301, USA.

TABLE 1. SILICATE WEATHERING FLUXES AND CO_2 CONSUMPTION IN PANAMA WATERHSEDS AND OTHER LOCALES

Geographic region*	n	Silicate weathering flux ($t\ km^{-2}\ yr^{-1}$)	CO_2 consumption ($mol \times 10^3\ km^{-2}\ yr^{-1}$)
This study			
Bocas Del Toro region	2	37.1–56.6	836–1274
Chiriquí region	11	30.6–69.5	688–1566
Veraguas-Cocles region	10	13.1–42.7	296–961
Azuero region	5	6.9–17.6	155–396
El Valle region	2	26.0–29.0	586–653
Chagres uplands	4	33.1–60.1	745–1161
Trans-Isthmus region	1	34.4	776
East Panama region	1	35.0	737
Bayano region	1	41.9	994
Andesitic-dacitic terrains			
Dominica[1]		6–106	190–1575
Martinique and Guadeloupe[2]		100–120	1100–1400
North Island, New Zealand[3, †]		15–310	217–2926
Basaltic-ultramafic terrains			
Deccan Traps[4]		21–63	580–2540
Phillipines[5]		114–396	233–6,190
Réunion Island[6]		63–170	1300–4400
Granitic terrains			
Nethravati River, India[7]		42	290
Puerto Rico[8]		37.8	

*Superscripts indicate references: 1—Goldsmith et. al. (2010); 2—Rad et. al. (2006); 3—Goldsmith et. al. (2008); 4—Dessert et. al. (2001); 5—Schopka et. al. (2011); 6—Louvat and Allègre (1997); 7—Gurumurthy et. al. (2012); 8—McDowell and Asbury (1994).
†Carbon export flux calculated as twice the silica flux.

TABLE 2. PEARSON CORRELATION COEFFICIENTS FOR WEATHERING FLUX ANALYSIS

Parameter	Silicate (sil) weathering flux (Ca_{sil}, Mg_{sil}, Na, K, Si)		
	n	r	*p* value
Runoff ($mm\ yr^{-1}$)	37	*0.998*	<0.0001
Physical erosion rates (m/m.y.)*	36	*0.756*	0.0028
Land cover (% forest)	34	*0.492*	0.003
Land cover (% agricultural)	34	0.342†	0.048
Land cover (percent grassland/shrubland)	33	*0.478†*	0.005
Elevation maximum (m)	36	*0.559*	0.0004
Maximum gradient	36	0.219	0.199
Mean gradient	36	*0.186*	0.277
Annual minimum mean precipitation (mm)	36	*0.590*	0.0002
Annual maximum mean precipitation (mm)	36	*0.584*	0.0002

Note: r is Pearson correlation coefficient. Values in italics indicate statistical significance.
*Data from Sosa-Gonzales (2011).
†Indicates negative correlation.

landscape denudation. Furthermore, our results add strong support to previous field-based studies that observed a strong correlation between chemical and physical erosion rates (Jacobson and Blum, 2003; Lyons et al., 2005), and are consistent with the observation by Johnsson (1990) of minimal chemical weathering in Panama riverbed sands. Overall, the chemical weathering yields for Panama rivers represents 3%–28% of the total denudation (i.e., physical + chemical denudation) within the sampled watersheds. This finding is supported by the significant positive correlation of weathering fluxes with mean stream gradient noted for the small mountainous rivers of New Zealand (Lyons et al., 2005).

In one of the first known attempts to relate chemical erosion fluxes to land cover and land-use practices in the humid tropics, we observe a slight positive, yet significant correlation with degree of watershed forest cover (Table 2). This positive correlation, coupled with negative correlations with amount of agricultural land and extent of grassland and shrubland cover, suggests that land-use practices have fundamentally altered flow paths and/or residence times of water within the critical zone, thus affecting watershed solute delivery potential across west-central Panama. This hypothesis is supported by initial hydrological findings from the Aqua Salud experimental watershed in Panama, where relatively higher dry season discharge values for forested watersheds compared to mixed use agricultural counterparts have been measured (Ogden et al., 2013). It is interesting that above-ground carbon density data for the Isthmus of Panama obtained via remote sensing by Asner et al. (2013) revealed that the highest values (approaching 130 Mg C/ha) corresponded well with our highest weathering fluxes. The fact that these locations are in the high mountains of the Cordillera Central, the Chagres uplands, and the Bayano region suggest an inherent link between high rainfall, chemical denudation potential, and amount of forest carbon stock in tropical landscapes.

The Panama silicate weathering fluxes and CO_2 consumption values from this study expand our understanding of the weathering potential of andesitic lithology, particularly in the humid tropics. Previous attempts to explore the CO_2 consumption potential of basaltic terrains in active margin settings revealed CO_2 drawdown values rates as high as 30%–35% of those for granitic continental terrains despite their limited aerial extent (Gaillardet et al., 1999; Dessert et al., 2003). A further exploration into controls of worldwide basalt weathering by Dessert et al. (2003) showed the importance of temperature and to a lesser extent runoff. Using their plot of atmospheric CO_2 consumption via silicate weathering, expressed as log HCO_3^- concentration versus $1000/T$ (K^{-1}) (temperature, in Kelvin) the average value of various andesite terrains is shown in Figure 2, compiling data from this study together with values from Martinique and Guadeloupe (Rad et al., 2006), Dominica (Goldsmith et al., 2010), and New Zealand (Goldsmith et al., 2008). These lower HCO_3^- values for andesite terrains compared to their basalt counterparts, even at these elevated

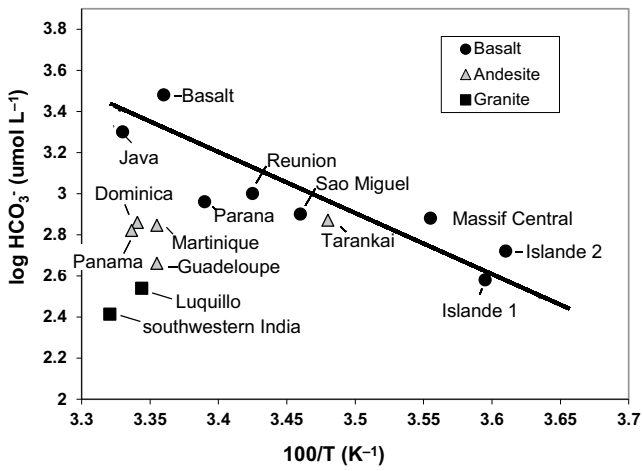

Figure 2. Plot of HCO_3^- versus $1000/T$ K^{-1} (temperature, in Kelvin) comparing CO_2 consumption in Panama with that for various lithologies globally (after Dessert et al., 2003; Goldsmith et al., 2010). All andesite values are below the trend line for basaltic terrains defined by Dessert et al. (2001), but above values for the granitic terrains of Luquillo (Puerto Rico; McDowell and Asbury, 1994) and southwestern India (Gurumurthy et al., 2012).

temperatures, appear to confirm an upper end limit for CO_2 drawdown potential for andesite weathering. However, the andesite values are distinctly higher than for granites from tropical locations in southwestern India (Gurumurthy et al., 2012) and Puerto Rico (McDowell and Asbury, 1994), among the highest known for this lithology. Therefore, while our results confirm the importance of andesite weathering as a CO_2 sink, they also point to the need for andesite to be considered separately from basalt and granite in modeling efforts that calculate modern worldwide CO_2 drawdown from silicate weathering.

CONCLUSIONS

A high-resolution, multiyear data set provides insight into controls on silicate weathering in tropical small mountainous rivers across west-central Panama. Strong statistical links of chemical weathering fluxes with runoff and precipitation, maximum elevation, and physical weathering rates confirm previous findings on the importance of runoff and mechanical denudation in maintaining elevated weathering rates on small mountainous rivers. The correlation of chemical weathering fluxes with forest land cover and forest carbon stock suggests a climatological and biological connection that warrants further investigation. The new Panama data contribute substantially to a growing body of literature suggesting that CO_2 drawdown from terrains with andesite lithology should be considered separately when calculating global CO_2 drawdown values from silicate weathering.

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